RETAIL STORE TRANSACTION

Data Analysis and Visualization — Semester 4th

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Subject	Data Analysis and Visualization
Semester	4th

Tools Used: Python | Pandas | Matplotlib

1. Introduction

This report presents a comprehensive analysis of a Retail Store Transaction dataset using Python. The analysis covers data loading and cleaning, exploratory data analysis (value counts), revenue breakdowns by location and product, payment type distribution, and store manager performance. All code is written using **Pandas** for data manipulation and **Matplotlib** for visualizations.

Total Transactions	199 rows
Stores	4 (Store A, B, C, D)
Products	7 (Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)
Store Managers	4 (Noah, Olivia, Mia, Liam)
Cashiers	5 (C1 – C5)
Payment Types	5 (Gift Card, Debit Card, Credit Card, Online, Cash)
Total Revenue	Rs.2,20,200.37

2. Data Loading and Cleaning

The dataset is loaded from an Excel file. Unnamed/empty columns are dropped and the first four rows are previewed to understand the structure.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("Retail-Store-Transactions.xlsx")

df.drop(columns=["Unnamed: 17"], inplace=True)
df = df.loc[:, ~df.columns.str.contains("^Unnamed")]

print(df.head(4))
```

time	StoreID	Location	Product	Qty	PaymentType	Cashier	TotalPrice
15:29	S3	Store C	Tablet	3	Gift Card	C1	1092.66
16:53	S1	Store B	Printer	9	Online	C5	3462.66
21:22	S9	Store B	Laptop	9	Debit Card	C3	3580.20
14:28	S9	Store C	Monitor	7	Online	C2	561.89

3. Exploratory Analysis — Value Counts

We analyse the frequency of each category column to understand the distribution of products, locations, managers, and cashiers.

```
print("Product counts:", df["Product"].value_counts())
```

```
print("Unique products:", df["Product"].unique())
print("Location counts:", df["Location"].value_counts())
print("Store Manager counts:", df["StoreManager"].value_counts())
print("Cashier counts:", df["Cashier"].value_counts())
```

Category	Values (count)
Products	Phone:38, Desk:31, Monitor:30, Printer:29, Laptop:28, Chair:25, Tablet:18
Locations	Store C:64, Store D:54, Store B:41, Store A:40
Store Managers	Noah:53, Olivia:53, Mia:47, Liam:46
Cashiers	C1:45, C5:42, C2:39, C4:38, C3:35

4. Revenue Analysis — By Location

Revenue is calculated per store using **groupby**. A bar chart visualises the comparison.

```
r_from_d = df[df["Location"] == "Store D"]["TotalPrice"].sum()
print("Revenue from Store D:", r_from_d) # 54490.08

r_each_l = df.groupby("Location")["TotalPrice"].sum()
print(r_each_l)

# Bar chart
r_each_l_graph = df.groupby("Location")["TotalPrice"].sum()
r_each_l_graph.plot(kind="bar")
```

Location	Revenue (Rs.)	Share (%)
Store A	36,979.24	16.8%
Store B	46,201.25	21.0%
Store C	82,529.80	37.5%
Store D	54,490.08	24.7%
TOTAL	**2,20,200.37**	**100%**

Insight: Store C is the top-performing location, generating 37.5% of total revenue.

5. Revenue Analysis — By Product

Revenue is calculated per product, both individually and via **groupby**.

```
tablet_rev = df[df["Product"]=="Tablet"]["TotalPrice"].sum()
printer_rev = df[df["Product"]=="Printer"]["TotalPrice"].sum()
phone_rev = df[df["Product"]=="Phone"]["TotalPrice"].sum()
# ... similar for Monitor, Laptop, Chair, Desk

r_each_p = df.groupby("Product")["TotalPrice"].sum()
print("Revenue per product:", r_each_p)
```

Product	Revenue (Rs.)	Rank
Phone	39,972.89	#1
Desk	39,519.04	#2
Monitor	39,149.59	#3
Printer	37,266.78	#4
Laptop	34,613.89	#5
Chair	15,524.54	#6
Tablet	14,153.64	#7

Insight: Phone generates the highest revenue; Tablet is the lowest.

6. Revenue — Location x Product Breakdown

A multi-level groupby reveals which product performs best in each store.

```
revenue = df.groupby(["Location", "Product"])["TotalPrice"].sum()
print("Revenue per location per product:", revenue)
```

Location / Product	Chair	Desk	Laptop	Monitor	Phone	Printer	Tablet
Store A	2,858	8,818	2,583	7,547	7,011	4,772	3,391
Store B	2,747	3,854	8,416	14,162	8,600	8,386	36
Store C	4,272	14,914	18,784	12,182	15,129	11,274	5,975
Store D	5,647	11,933	4,832	5,259	9,233	12,835	4,752

7. Payment Type Analysis

We examine which payment methods are most frequently used and their revenue contribution per store.

```
print("Payment types:", df["PaymentType"].unique())
print("Count per type:", df["PaymentType"].value_counts())

loc_pay = df.groupby(["Location","PaymentType"])["TotalPrice"].sum()
print(loc_pay)

loc_pay_cnt = df.groupby(["Location","PaymentType"]).size().unstack()
loc_pay_cnt.plot(kind="bar")
plt.ylabel("Count of Payment Type")
plt.title("Payment Type Count per Store")
plt.show()
```

Payment Type	Count	Share (%)
Gift Card	49	24.6%
Debit Card	43	21.6%
Credit Card	43	21.6%

Online	35	17.6%
Cash	29	14.6%

Insight: Gift Card is the most popular payment method (24.6%), while Cash is the least used.

8. Store Manager Analysis

We assess each store manager's transaction count, total revenue, average sale, and cashier team distribution.

```
storemanager_names = df["StoreManager"].unique()
print("NUMBER OF STORE MANAGERS:", len(storemanager_names))

val_of_each = df["StoreManager"].value_counts()
print("Transactions per manager:", val_of_each)

grp_cashier_mgr = df.groupby(["Cashier","StoreManager"]).size()
print(grp_cashier_mgr)

# Bar: cashier vs manager
grp_cashier_mgr.unstack().plot(kind="bar")
plt.title("Cashier group with manager")
plt.show()
```

Manager	Transactions	Total Revenue (Rs.)	Avg Sale (Rs.)
Noah	53	59,356.69	1,119.94
Olivia	53	53,574.86	1,010.85
Mia	47	55,876.13	1,188.85
Liam	46	51,392.69	1,117.23

Insight: Noah and Olivia lead in transaction count (53 each); Mia has the highest average sale.

9. Pivot Table — Manager x Product Revenue

A pivot table cross-tabulates each manager's revenue broken down by product category, providing a comprehensive performance matrix.

```
summary = df.groupby("StoreManager").agg(
Total_Sales=("TotalPrice", "sum"),
Avg_Sale=("TotalPrice", "mean"),
Number_of_Transactions=("TotalPrice", "count")
)

pivot = df.pivot_table(
index="StoreManager",
columns="Product",
values="TotalPrice",
aggfunc="sum",
margins=True,
```

```
margins_name="Total_All_Products"
)
print(summary)
print(pivot)
```

Manager	Chair	Desk	Laptop	Monitor	Phone	Printer	Tablet	TOTAL
Liam	2,075	8,691	10,078	11,082	5,905	10,597	2,965	51,393
Mia	894	12,099	10,888	6,583	11,789	8,009	5,614	55,876
Noah	6,266	12,241	10,751	7,820	9,718	8,645	3,916	59,357
Olivia	6,290	6,488	2,897	13,665	12,562	10,015	1,659	53,575
TOTAL	**15,525**	**39,519**	**34,614**	**39,150**	**39,973**	**37,267**	**14,154**	**2,20,200**

10. Key Insights and Conclusion

- **Store C is the revenue leader**
  Store C contributed Rs.82,530 (37.5%) of total revenue, nearly double Store A's contribution.
- **Phone drives the most revenue**
  Phone generated Rs.39,973 — the highest among all 7 products. Tablet had the lowest at Rs.14,154.
- **Gift Card dominates payments**
  49 out of 199 transactions used Gift Card, making it the most preferred payment method.
- **Noah leads in transactions**
  Manager Noah handled 53 transactions (tied with Olivia), but Mia achieved the highest average sale of Rs.1,188 per transaction.
- **Store B excels in Monitor sales**
  Store B recorded Rs.14,162 from Monitor alone — the single highest product-location combination.
- **Store C x Laptop is the top combo**
  Store C generated Rs.18,784 from Laptop sales, the highest product-location revenue pair.

This analysis demonstrates how Python's Pandas and Matplotlib libraries can efficiently extract actionable insights from retail transaction data — covering revenue trends, product performance, payment preferences, and manager effectiveness.